



SRI SATHYA SAI INSTITUTE OF HIGHER LEARNING
(Deemed to be University)
Department of Mathematics and Computer Science

CIE-III ASSIGNMENT

UMAM-502: Optimization Techniques
for Machine Learning

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Date: 12-09-2026

- Consider the hyperplane $H = \{x \in R^3 : 3x_1 + 2x_2 - x_3 = 6\}$ and the point $y = (1, 2, 3)^T$. [Unit 1 | BTL-3]
 - Compute the distance from y to H and the orthogonal projection of y onto H .
 - Show that the half-space $H_- = \{x : 3x_1 + 2x_2 - x_3 \leq 6\}$ is a convex set.
- Let $g: R^3 \rightarrow R^2$ be $g(x) = (x_1^2 + x_2x_3, x_1x_2 - x_3^2)^T$ and let $f(x) = x_1^2 + x_2x_3$. Take $x^0 = (1, 2, -1)^T$. [Unit 1 | BTL-3]
 - Find the derivative (Jacobian) matrix $Dg(x^0)$.
 - Find the level set of f through x^0 , the tangent plane to it at x^0 , and the direction of steepest increase of f at x^0 .
- For $f(x_1, x_2) = x_1^3 + 3x_1x_2^2$, construct the second-order Taylor approximation about $x^0 = (1, 1)^T$. Use it to estimate $f(1.1, 0.9)$, compare with the exact value, and **explain** why the quadratic correction term contributes nothing at this particular point. [Unit 1 | BTL-4]
- Minimize $f(x) = x^4 - 14x^3 + 60x^2 - 70x$ over the initial interval $[0, 2]$ using the **golden section** method. Perform two iterations and state the final interval of uncertainty and its reduction factor. [Unit 2 | BTL-3]

5. Minimize $f(x) = x^3 - 3x + 1$ over $[0, 2]$ using the **Fibonacci method** with $N = 4$ iterations and $\varepsilon = 0.05$. Verify the final reduction factor against the theoretical value. **[Unit 2 | BTL-3]**

6. For $f(x) = x_1^2 + 2x_2^2$ with $x^{(0)} = (2, 1)^T$: **[Unit 2 | BTL-4]**

- Perform one iteration of steepest descent with **exact** line search.
- Perform one iteration of **Newton's method**.
- Compare** the two and verify the observed decrease against the theoretical convergence rate.

7. Solve by the simplex method:

$$\text{Maximize } z = 3x_1 + 5x_2 \quad \text{s.t.} \quad x_1 \leq 4, \quad 2x_2 \leq 12, \quad 3x_1 + 2x_2 \leq 18, \quad x_1, x_2 \geq 0$$

[Unit 3 | BTL-3]

8. For the LP of Question 7 and its optimal tableau: **[Unit 3 | BTL-4]**

- Write the dual and read its optimal solution from the final tableau; verify strong duality.
- Interpret** the shadow prices and determine the range of $b_2 = 12$ over which the current basis remains optimal, together with the effect on z^* .

9. Minimize $f(x) = x_1^2 + x_2^2 + x_3^2$ subject to

$$h_1(x) = x_1 + x_2 + x_3 - 3 = 0, \quad h_2(x) = x_1 - x_2 + 2x_3 - 4 = 0$$

Apply the Lagrange condition to find the candidate point and verify the second-order sufficiency condition. **[Unit 4 | BTL-3]**

10. Minimize $f(x) = (x_1 - 2)^2 + (x_2 - 1)^2$ subject to $g_1(x) = x_1 + x_2 - 2 \leq 0$, $g_2(x) = -x_1 \leq 0$, $g_3(x) = -x_2 \leq 0$. **Analyse** all activity cases using the KKT conditions, obtain the solution, and confirm it with the second-order condition. **[Unit 4 | BTL-4]**
